

Georgios Pantazopoulos

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Education

PhD in Robotics and Autonomous Systems

Sep 2021 - July 2026

CDT in Robotics and Autonomous Systems, University of Edinburgh, Heriot-Watt University

- Thesis: From Pixels to Actions: Grounding in Vision-Language Models, supervised by [A. Suglia](#)

MSc in Artificial Intelligence with Speech and Multimodal Interaction

Sep 2020 - Aug 2021

Heriot-Watt University

- Grade: 84/100 (1st with Distinction)
- Thesis: Compositional Generalization in Image Captioning, supervised by [A. Eshghi](#)

MEng in Electrical and Computer Engineering

Sep 2012 - Jun 2018

National Technical University of Athens

- Grade: 7.7/10, Major: Computer Science
- Thesis: Stress Detection from Speech Signals, supervised by [A. Potamianos](#)

Publications

Retrievit: In-context Retrieval Capabilities of Transformers, State Space Models, and Hybrid Architectures

2026 *Transactions on Machine Learning Research*, [\[paper\]](#)

- G. Pantazopoulos, M. Nikandrou, I. Konstas, A. Suglia

An Efficient Training Pipeline for Reasoning Graphical User Interface Agents

2026 *ICLR 2026 Workshop on Multimodal Intelligence*, [\[paper\]](#)

- G. Pantazopoulos, E. Özyiğit

Evaluating Multimodal Language Models as Visual Assistants for Visually Impaired Users

2025 *Annual Meeting of the Association for Computational Linguistics*, [\[paper\]](#)

- A. Karamolegkou*, M. Nikandrou*, G. Pantazopoulos*, D. Sanchez, P. Rust, R. Dhar, D. Hershcovich, A. Søgaard

CROPE: Evaluating In-Context Adaptation of Vision and Language Models to Culture-Specific Concepts

2025 *Annual Conference of the North American Chapter of the Association for Computational Linguistics*, [\[paper\]](#)

- M. Nikandrou, G. Pantazopoulos, N. Vitsakis, I. Konstas, A. Suglia

Shaking Up VLMs: Comparing Transformers and Structured State Space Models for Vision & Language Modeling

2024 *Conference on Empirical Methods in Natural Language Processing*, [\[paper\]](#)

- G. Pantazopoulos, M. Nikandrou, A. Suglia, O. Lemon, A. Eshghi

Enhancing Continual Learning in Visual Question Answering with Modality-Aware Feature Distillation

2024 *ACL Third Workshop on Advances in Language and Vision Research*, [\[paper\]](#)

- M. Nikandrou*, G. Pantazopoulos*, I. Konstas, A. Suglia

Lost in Space: Probing Fine-grained Spatial Understanding in Vision and Language Resamplers

2024 *Annual Conference of the North American Chapter of the Association for Computational Linguistics*, [\[paper\]](#)

- G. Pantazopoulos, A. Suglia, O. Lemon, A. Eshghi

Learning To See But Forgetting To Follow: Visual Instruction Tuning Makes LLMs More Prone To Jailbreak Attacks

2024 *LREC-COLING Third Workshop on Safety for Conversational AI*, [\[paper\]](#)

- G. Pantazopoulos, A. Parekh, Malvina Nikandrou, A. Suglia

Multitask Multimodal Prompted Training for Interactive Embodied Task Completion

2023 *Conference on Empirical Methods in Natural Language Processing*, [\[paper\]](#)

- G. Pantazopoulos, M. Nikandrou, A. Parekh, B. Hemanthage, A. Eshghi, I. Konstas, V. Rieser, O. Lemon, A. Suglia

EMMA: A foundation model for embodied, interactive, multimodal task completion in 3D environments

2023 *Alexa Prize SimBot Challenge Proceedings*, [\[paper\]](#)

- A. Parekh*, M. Nikandrou*, G. Pantazopoulos*, B. Hemanthage, A. Eshghi, I. Konstas, O. Lemon, A. Suglia

Combine to Describe: Evaluating Compositional Generalization in Image Captioning

2022 *ACL Student Research Workshop*, [\[paper\]](#)

- **G. Pantazopoulos**, A. Suglia, A. Eshghi

Data Augmentation using GANs for Speech Emotion Recognition

2019 Annual Conference of the International Speech Communication Association, [\[paper\]](#)

- A. Chatziagapi, G. Paraskevopoulos, D. Sgouropoulos, **G. Pantazopoulos**, M. Nikandrou, T. Giannakopoulos, A. Katsamanis, A. Potamianos, S. Narayanan

Projects

Retrievit

Dec 2025 - Mar 2026

University of Edinburgh, [\[code\]](#)

- Trained and tested Transformers, SSMs, and Hybrid interleaved/dual stream models for in-context retrieval
- Demonstrated that hybrids excel in length generalization while matching Transformers efficiency
- Conducted representational analysis in embedding spaces showing locality-preserving structures in SSMs/hybrids

VLMs as Visual Assistants

Dec 2024 - Jan 2025

Heriot-Watt University, [\[code\]](#)

- Showed that current VLMs are not successfully responding to Braille text from images
- Developed a dataset for Braille recognition by rendering text in images
- Trained models achieving more than 10-fold performance improvement in cross-script QA

Mamba-VL

Dec 2023 - Mar 2024

Heriot-Watt University, [\[code\]](#)

- Trained & Benchmarked 3 large-scale VLMs using Transformer and Mamba-based LLMs
- Benchmarked across 13 vision-language tasks spanning varying levels of perceptual granularity
- Demonstrated superior captioning, VQA, and reading-comprehension performance
- Used synthetic associative recall tasks to measure performance trade-offs

EMMA

Jan 2022 - May 2023

Alexa Prize Simbot Challenge, Finalist [\[code\]](#)

- Co-led the pre-training of the multimodal agent achieving state-of-the-art results in image understanding
- Fine-tuned the base model for hierarchical decision making enabling tool and boosting success rate by 7%
- Contributed to self-play synthetic data generation pipelines doubling task success

Exploring the Role of Patch Resolution for Vision Transformers

Jan 2022 - Mar 2022

University of Edinburgh

- Pretrained vision transformers from scratch
- Demonstrated that ViTs have sharp attention score distributions in deeper layers
- Replaced self with linear attention achieving $2.3 \times$ inference speed with minimal performance loss

Work Experience

AI Research Intern

Jun 2025 - Dec 2025

Alan Turing Institute

- Trained GUI agents with, supervised fine-tuning, chain-of-thought training, and reinforcement learning
- Developed data cleaning pipelines achieving 400% compression ratio with no performance degradation
- Trained ranker models to detect false-positive across 4.8M examples improving F1 by 13%

AI Engineer Intern

Feb 2025 - Apr 2025

Reka AI

- Developed contrastive models improving recall by 15% in image-to-image/video & text-to-video retrieval
- Contributed to search-augmented VLMs with embedding models for video retrieval
- Implemented instance retrieval from natural language descriptions in videos

Machine Learning Engineer

Sep 2018 - Apr 2020

Behavioral Signal Technologies Inc.

- Developed CNN-based classification models for emotion recognition from raw audio signals
- Built LSTM-based classification models for emotion recognition from automatic transcriptions
- Designed late/early-fusion methods for audio and text improving performance by 8% over unimodal models
- Led the data collection from crowdsourcing platforms resulting in the collection of 600k annotated examples

Teaching Experience

A@NLP 2024 / 2025

Sep 2024 / Sep 2025

Summer School Teaching Assistant, [\[code\]](#)

- Developed the lab materials for sequence modeling and multimodal models

Data Mining and Machine Learning
Teaching Assistant

Sep 2022 - present

- Contributed to the infrastructure and the coursework of the module
- Supervised groups of postgraduate students on their semester project

Introduction to Natural Language Processing
Teaching Assistant, [\[code\]](#)

Sep 2023 - present

- Contributed to the infrastructure and the coursework of the module
- Supervised groups of students on their semester project

Technical skills

Programming Languages	Python, Matlab, R, C, C++, C#, Java, PDDL, Bash, SML
ML Frameworks	PyTorch, Hugging Face, Transformers, PEFT, Accelerate, TRL, PyTorch Lightning, scikit-learn, spacy, openCV, Albumentations, YOLO, Gradio
Data Engineering	Pydantic, Pandas, PyArrow
MLOPs & CI/CD	Weights & Biases, MLflow, Github Actions
Vector Databases	Faiss, Milvus, chroma, Weaviate
Training & Inference	Slurm, Docker, Kubernetes, vLLM
Evaluation & Benchmarking	large-scale data collection & filtering, data and model ablations

Honors & Awards

Evaluating Multimodal Language Models as Visual Assistants for Visually Impaired Users <i>Pioneer Center of AI Best Paper Award, ACL 2025 SAC Highlights</i>	Jul 2025
Exploring the Role of Patch Resolution for Vision Transformers <i>University of Edinburgh, 2nd place in the annual IBM Awards</i>	Mar 2022

Invited Talks

An Efficient Training Pipeline for Reasoning Graphical User Interface Agents <i>Alan Turing Institute</i>	Nov 2025
Embodied MultiModal Agent for Language-guided Action Execution in 3D Simulated Environments <i>CDT in Robotics and Autonomous System Conference, [slides]</i>	Oct 2022

Academic Service

COLM Reviewer	May 2026
ICLR Reviewer	Jan 2026
ARR Reviewer	Jun 2024 / Oct 2024 / Jun 2025
IEEE SLT218 Conference Volunteer	Dec 2018